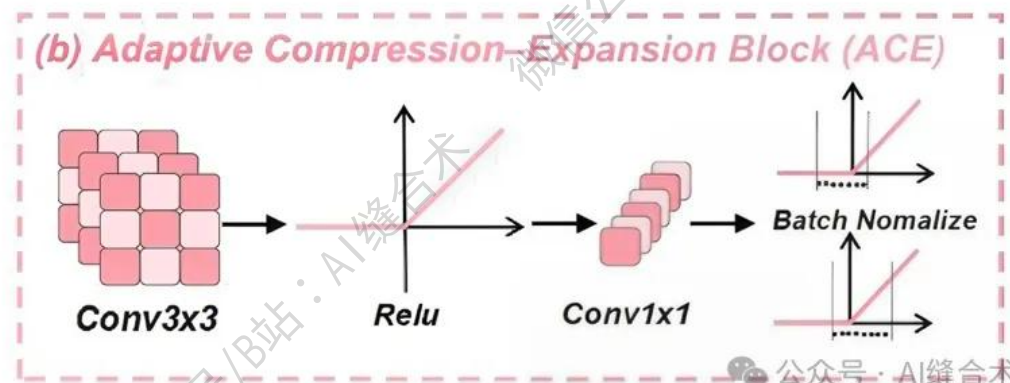


核心速览



论文题目：SEMC: Structure-Enhanced Mixture-of-Experts Contrastive Learning for Ultrasound Standard Plane Recognition

中文题目：SEMC ：结构增强的专家混合对比学习用于超声标准平面识别

论文链接：<https://arxiv.org/pdf/2511.12559>

所属单位：中国海洋大学信息科学与工程学院，香港理工大学健康科技与信息学系，山东大学信息科学与工程学院

核心速览：提出 SEMC 框架，通过语义-结构融合模块和混合专家对比识别模块，结合多尺度结构信息与专家引导对比学习，提升超声标准平面识别性能，并构建 LP2025 肝脏超声数据集。

论文内容详细解读：

<https://mp.weixin.qq.com/s/qWYfaTeBmQ3KbmfQFWUDuQ>

down_times=1

```
ACE(
  (blocks): Sequential(
    (0): Sequential(
      (0): Conv2d(3, 3, kernel_size=(3, 3), stride=(2, 2), padding=(1, 1), groups=3, bias=False)
      (1): BatchNorm2d(3, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      (2): ReLU(inplace=True)
      (3): Conv2d(3, 64, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (4): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      (5): ReLU(inplace=True)
    )
  )
  (out_proj): Sequential(
    (0): Conv2d(64, 64, kernel_size=(1, 1), stride=(1, 1), bias=False)
    (1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  )
)
input_tensor_shape : torch.Size([1, 3, 256, 256])
output_tensor_shape : torch.Size([1, 64, 128, 128])
```

哔哩哔哩/微信公众号：AI缝合术，独家整理！

down_times=2

```
ACE(
  (blocks): Sequential(
    (0): Sequential(
      (0): Conv2d(3, 3, kernel_size=(3, 3), stride=(2, 2), padding=(1, 1), groups=3, bias=False)
      (1): BatchNorm2d(3, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      (2): ReLU(inplace=True)
      (3): Conv2d(3, 6, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (4): BatchNorm2d(6, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      (5): ReLU(inplace=True)
    )
    (1): Sequential(
      (0): Conv2d(6, 6, kernel_size=(3, 3), stride=(2, 2), padding=(1, 1), groups=6, bias=False)
      (1): BatchNorm2d(6, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      (2): ReLU(inplace=True)
      (3): Conv2d(6, 64, kernel_size=(1, 1), stride=(1, 1), bias=False)
      (4): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      (5): ReLU(inplace=True)
    )
  )
  (out_proj): Sequential(
    (0): Conv2d(64, 64, kernel_size=(1, 1), stride=(1, 1), bias=False)
    (1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  )
)
input_tensor_shape : torch.Size([1, 3, 256, 256])
output_tensor_shape : torch.Size([1, 64, 64, 64])
```